End-TermExamination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech: IT/CSE-AI>Semester:< 5 >
(Nov-Dec, 2022)

Subject Code:BCS 303 Subject: Theory of Computation
Time:3 Hours Maximum Marks :60
Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (5*4=20)

(a) Construct a DFA that accepts the language L= $\{a^n \mid n$ is either a multiple of 3 or a multiple of 5$\}$.

(b) List and explain the types of grammars with example according to Chomsky classification?

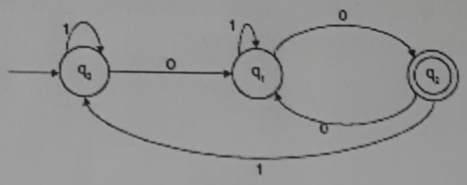
(c) Compare and contrast finite automaton, Push Down Automaton and Turing machines.

(d) Write the grammar for the language L= $\{a^n b^n c^m d^m \mid n,m \geq 1\}$.

UNIT-I

Q2 (a) Construct finite automata which will accept string of 0's and 1's containing even number of 0's and odd number of 1's. (5*2=10)

(b) Construct a Regular expression from the given Finite Automata using Arden's theorem



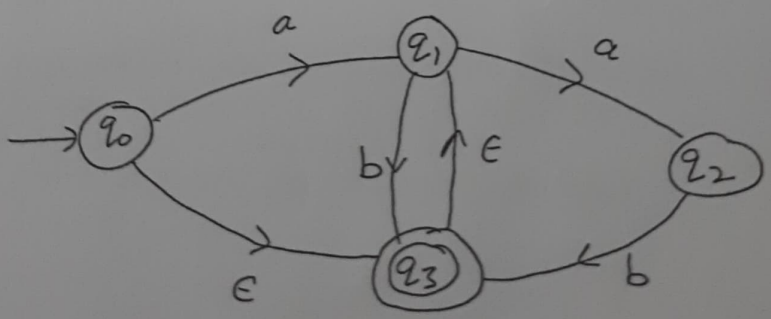
Q3 Design an optimized DFA equivalent to NFA (10)
M = ({p, q, r, s}, {0, 1}, δ, p, {q, s})

Where δ is given below:

δ(p,0) = q	δ(r, 0) = s
δ(p, 1) = (q, s)	δ(r,1) = φ
δ(q, 0) = q	δ(s, 0) = p
δ(q, 1) = (p, r)	δ(s, 1) = q

UNIT-II

Q4 Why is an FA with ϵ transition called NFA? What is the necessity of NFA with ϵ transition? Convert the following ϵ-NFA to an equivalent optimizedDFA. (10)



PTO

Q5 What is the use and significance of Normal forms? Differentiate between Chomsky normal form and Greibach Normal form. Reduce the following CFG into Chomsky normal form (10)

$$S \rightarrow aSb \mid aA \mid bB \mid a \mid b, \quad A \rightarrow aA \mid a, B \rightarrow bB \mid b$$

UNIT-III

Q6 Design a Push down Automata that accepts the language $L = \{a^n b^{2n} \mid n > 0\}$ and show the Instantaneous Description for the string "aabbbb". (10)

Q7 (a) Simplify the following CFG. (5*2=10)

$$S \rightarrow AaB \mid aaB,\ A \rightarrow D,\ B \rightarrow bbA \mid \epsilon, D \rightarrow E,\ E \rightarrow F,\ F \rightarrow aS$$

(b) Let G be a grammar
$S \rightarrow aSX \mid b,\ X \rightarrow Xb \mid a$
For the String'aababa' find the leftmost derivation, rightmost derivation, and parse tree.

UNIT-IV

Q8 Design a Turing Machine that accepts the language $L = \{a^n b^n c^n \mid n \geq 1\}$ and show the Instantaneous Description for the string "aabbcc". (10)

Q9 Design a Moore machine, which takes as input a binary stream and generates an output of 1 only when a sequence of the pattern 01011 is found in the input stream.Convert the Moore machine that you have got into a Mealy machine.Check whether the mealy machine you obtained is a minimal one or not. Give proper justification to your answer. (10)